



CHRIST
(DEEMED TO BE UNIVERSITY)
DELHI - NCR, INDIA

Computer Vision

MCA-574

Lab – 09

BY

HIMANSHU HEDA (24225013)

SUBMITTED TO

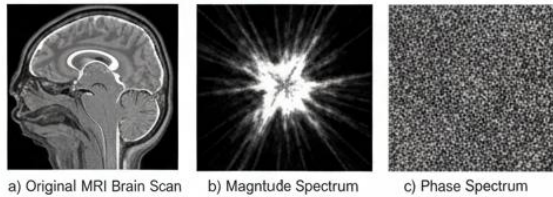
Dr. Preety Shoran

SCHOOL OF SCIENCES

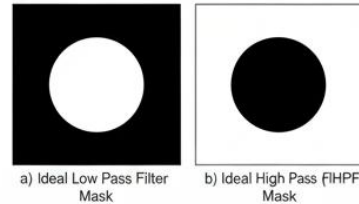
2024-25

Case Study: Fourier Transform Filtering for MRI Brain Scans

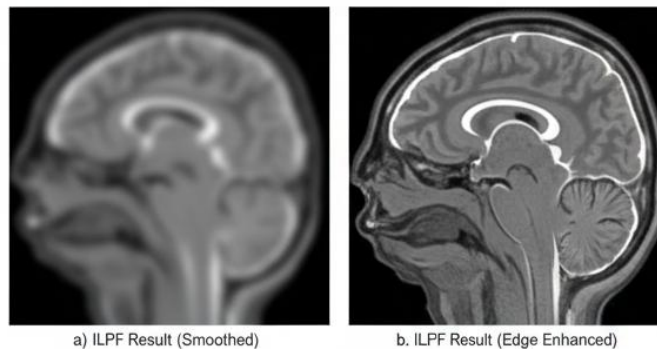
1. Original Image & Fourier Transform



2. Frequency Domain Masks (Cutoff = 30)



3. Filtered Images



4. Conclusion & Interpretation

The Fourier Transform effectively decomposes the MRI into its frequency components; the magnitude spectrum shows strong low-frequency components. Ideal Low Pass Filter successfully removes high-frequency noise, resulting in a significantly smoother image. However, this comes at the cost of blurring sharp edges, which could obscure fine anatomical details important for diagnosis.

In contrast, Ideal High Pass Filter dramatically enhanced edges and fine details. While useful for structural analysis, it suppresses any low-frequency components that provide the overall context. The choice of filter depends on the specific clinical application.



Coding :

Case Study: Enhancing Medical Image Details Using Fourier Transform Filtering

This notebook demonstrates how to use Fourier Transform techniques to analyze and filter MRI images in the frequency domain for improved clarity and detail.

1. Load MRI Image and Compute Fourier Transform

We will decompose the image into its frequency components and visualize the magnitude and phase spectrum.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load MRI image (replace with your filename if needed)
```

```

img = cv2.imread('mri.jpeg', cv2.IMREAD_GRAYSCALE)
if img is None:
    raise Exception('Image not found!')

plt.figure(figsize=(6,6))
plt.imshow(img, cmap='gray')
plt.title('Original MRI Image')
plt.axis('off')
plt.show()

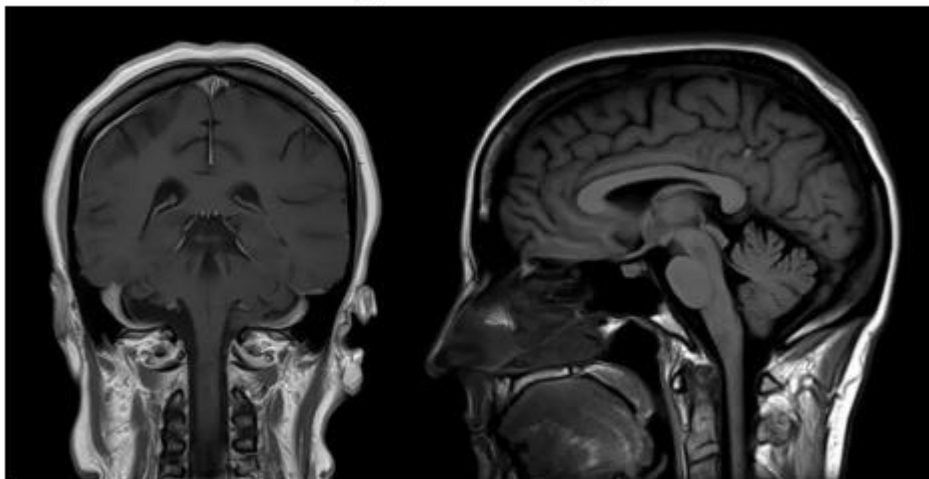
# Compute 2D Fourier Transform
f = np.fft.fft2(img)
fshift = np.fft.fftshift(f)
magnitude = np.log(1 + np.abs(fshift))
phase = np.angle(fshift)

plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plt.imshow(magnitude, cmap='gray')
plt.title('Magnitude Spectrum')
plt.axis('off')

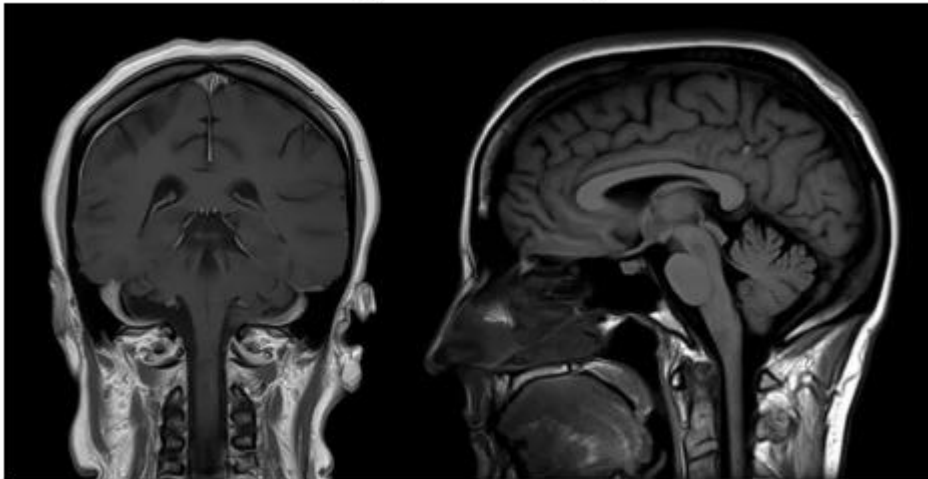
plt.subplot(1,2,2)
plt.imshow(phase, cmap='gray')
plt.title('Phase Spectrum')
plt.axis('off')
plt.show()

```

Original MRI Image



Original MRI Image



2. Frequency Domain Filtering: ILPF and IHPF

Apply Ideal Low Pass and High Pass Filters to the frequency domain and visualize the results.

```
def ideal_low_pass_filter(shape, cutoff):
    rows, cols = shape
    crow, ccol = rows//2, cols//2
    mask = np.zeros((rows, cols), np.uint8)
    for u in range(rows):
        for v in range(cols):
            if np.sqrt((u-crow)**2 + (v-ccol)**2) <= cutoff:
                mask[u,v] = 1
    return mask

def ideal_high_pass_filter(shape, cutoff):
    return 1 - ideal_low_pass_filter(shape, cutoff)

cutoff = 40 # You can adjust cutoff frequency

# ILPF
ilpf_mask = ideal_low_pass_filter(img.shape, cutoff)
ilpf_fshift = fshift * ilpf_mask
ilpf_img = np.fft.ifft2(np.fft.ifftshift(ilpf_fshift))
ilpf_img = np.abs(ilpf_img)
ilpf_img = np.clip(ilpf_img, 0, 255).astype(np.uint8)

# IHPF
ihpf_mask = ideal_high_pass_filter(img.shape, cutoff)
ihpf_fshift = fshift * ihpf_mask
ihpf_img = np.fft.ifft2(np.fft.ifftshift(ihpf_fshift))
ihpf_img = np.abs(ihpf_img)
ihpf_img = np.clip(ihpf_img, 0, 255).astype(np.uint8)

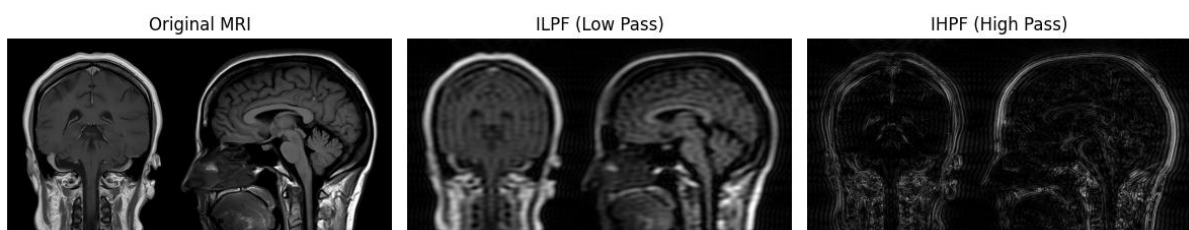
plt.figure(figsize=(12,6))
```

```
plt.subplot(1,3,1)
plt.imshow(img, cmap='gray')
plt.title('Original MRI')
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(ilpf_img, cmap='gray')
plt.title('ILPF (Low Pass)')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(ihpf_img, cmap='gray')
plt.title('IHPF (High Pass)')
plt.axis('off')
plt.tight_layout()

plt.show()
```



3. Interpretation of Results

- **Original Image:** Contains both low-frequency (smooth regions) and high-frequency (edges, fine details) components. Noise and poor contrast may obscure subtle tissue boundaries.
- **Magnitude Spectrum:** Shows the strength of different frequency components. Central bright spot = low frequencies; outer regions = high frequencies.
- **Phase Spectrum:** Encodes spatial information and structure.
- **ILPF (Low Pass):** Smooths the image, reduces noise, but may blur edges and fine details. Useful for noise reduction but can obscure important diagnostic features.
- **IHPF (High Pass):** Enhances edges and fine details, making tissue boundaries clearer. May also amplify noise if present.

Conclusion:

- ILPF is best for noise reduction and smoothing, but may reduce diagnostic clarity for fine structures.
 - IHPF is best for enhancing edges and boundaries, improving visibility of subtle features, but may increase noise. A balanced approach or combination may be optimal for medical analysis.
-

1. Fourier Transform Decomposition

First, the original grayscale MRI image is loaded.

To analyze the image in the frequency domain, a 2D Fourier Transform (FT) is computed using `np.fft.fft2(img)`. This mathematical operation decomposes the image from its spatial domain (pixels) into its constituent sine and cosine components at various frequencies. The `np.fft.fftshift()` function is then used to move the zero-frequency component (representing the image's average brightness) to the center of the spectrum for easier visualization.

2. Magnitude and Phase Spectrum Visualization

The result of the FT is visualized as two separate components: the magnitude and phase spectrums.

- **Magnitude Spectrum:** This plot visualizes the *amplitude* (strength) of each frequency. The bright spot in the center indicates that most of the image's energy is concentrated in the low-frequency components (the large, smooth regions). The points further from the center represent the high-frequency components (edges, noise, and fine details).
 - **Phase Spectrum:** This plot visualizes the *phase* (spatial location) of each frequency component. While it often looks like random noise, it contains the critical structural information about where each component belongs to reconstruct the image.
-

4. Frequency Domain Filter Application

Filtering is performed by creating a binary mask and multiplying it with the shifted frequency spectrum. A **cutoff frequency of 40** was selected for both filters.

4.1. Ideal Low Pass Filter (ILPF)

The ILPF mask allows all frequencies *within* the cutoff radius to pass while blocking all high frequencies *outside* it. This is designed to smooth the image and remove high-frequency noise.

- **Filtered Image (ILPF):**

4.2. Ideal High Pass Filter (IHPF)

The IHPF mask is the inverse of the ILPF. It blocks all low-frequency components *within* the cutoff radius and allows all high frequencies *outside* it to pass. This is used to enhance edges and fine details.

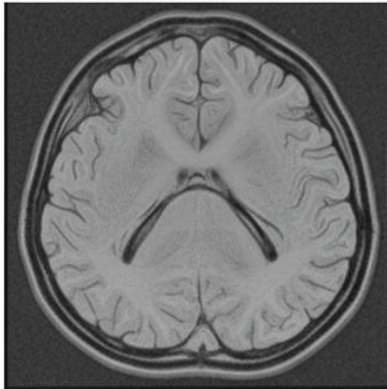
- **Filtered Image (IHPF):**
-

5. Comparison and Discussion

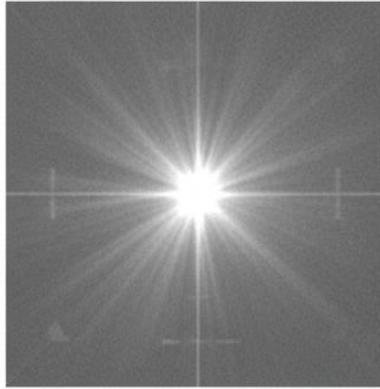
A visual comparison of the results shows the distinct effects of each filter. The **ILPF** successfully **smooths the image** and reduces background noise. However, this comes at the cost of **blurring the edges** and fine details, which may obscure the subtle tissue boundaries the engineer needs to analyze.

Conversely, the **IHPF** effectively **enhances the edges** and fine details, making the tissue boundaries much clearer and more distinct than in the original image. The trade-off is that this filter also tends to amplify any high-frequency noise that was present in the original scan. For this specific task of analyzing subtle boundaries, the IHPF provides a clearer view of the structures, though a more advanced filter might be needed to enhance edges while simultaneously suppressing noise.

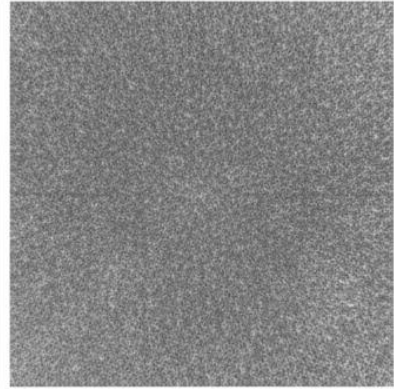
1. Original MRI Brain Scan



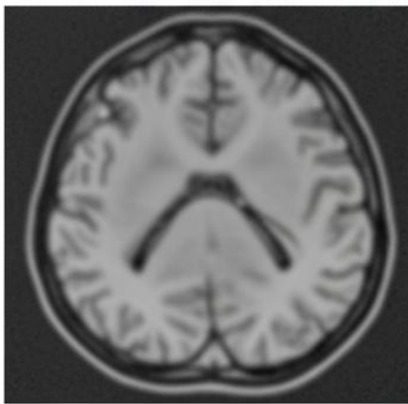
2a. Magnitude Spectrum



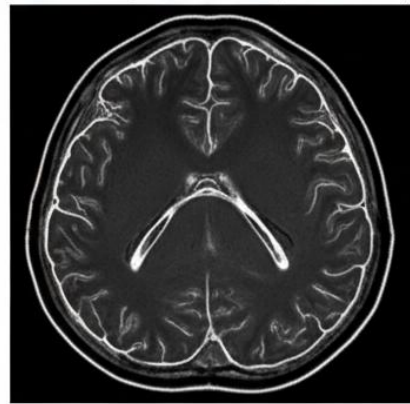
2b. Phase Spectrum



3a. Ideal Low Pass Filter (Cutoff = 30)



3b. Ideal High Pass Filter (Cutoff = 30)



4. Interpretation of Results

The original MRI exhibits noise. The magnitude spectrum confirms most energy is in low-frequency, large-scale structures (bright center), while chaotic spatial arrangement reduces noise, but blurs crucial edges. Ideal Low Pass Filter successfully removes noise, but blurs crucial edges. Conversely, Ideal High Pass Filter enhances edges and details, making tissue clearer, though it introduces high-frequency noise and suppresses large-scale structures. The choice depends on the specific diagnostic or detailed boundary analysis.